

Technical Signal Discovery in Daily OHLCV Data

A Walk-Forward Comparison with Reversal and Momentum

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Abstract

We compare simple reversal and momentum signals with a broad library of OHLCV technical features for U.S. equities. Four elastic-net specifications are evaluated at one- and five-day horizons in yearly walk-forward tests from 2013 through 2025. The baseline and its interaction extension attain the largest average rank correlations, while the technical specifications produce greater separation between the extreme deciles. Pruning 103 covered features to 85 has little effect on out-of-sample results. Predictability is small, varies materially across years, and is reported before transaction costs.

1 Introduction

Daily prices and volume support a large number of technical signals. Many are different transformations of the same recent path, so a large library can contain more redundancy than independent information. Kakushadze’s formulaic alphas make this issue explicit: the formulas are diverse in appearance, yet often combine the same primitives—returns, ranges, volume, ranks, delays, and rolling correlations [3]. This study asks how broad technical specifications compare with a simple three-signal benchmark at one- and five-day horizons. The benchmark contains one-day reversal, five-day reversal, and 12–1 momentum. The technical library contains 90 implementable formulaic alphas and 15 conventional OHLCV signals. We compare the benchmark, a benchmark with pairwise interactions, a target-blind pruned library, and the full covered library. All four specifications use the same elastic-net estimator and yearly walk-forward design.

The comparison separates two questions. Information coefficient (IC) measures the quality of the full cross-sectional ranking. A decile portfolio measures whether the ranking is useful at the two tails. The experiments show that these criteria need not agree. This difference, together with the stability of the selected coefficients across years, is the main empirical result.

2 Data and Signal Construction

2.1 Panel and prediction target

The course OHLCV panel contains the largest 1,000 U.S. equities by month-end market capitalization from February 2001 through December 2025. Securities are identified by CRSP `permno`. After removing rows with a missing close or market capitalization, the sample contains 6.25 million security-days and 3,253 securities. The daily cross-section has a median of 998 names.

Signals use information dated t to predict the split-adjusted close-to-close price return

$$y_{i,t}^{(h)} = \frac{C_{i,t+h}}{C_{i,t}} - 1, \quad h \in \{1, 5\}. \quad (1)$$

The panel’s CRSP `ret` field is backward-looking and includes distributions. A separate audit finds that `ret` and the corresponding close-to-close return differ by less than one basis point on 99.0% of

observations. Equation (1) therefore measures price appreciation rather than total return. This differs from the project brief’s preferred construction from compounded `ret`; the implications are discussed in Section 5.1.

Date gaps longer than ten calendar days start a new listing segment. Targets and rolling features are set to missing whenever their windows cross a segment boundary. This prevents an exit and later re-entry from being treated as adjacent trading days. A true VWAP field is unavailable, so formulas requiring VWAP use the daily typical price $(H + L + C)/3$.

2.2 Baseline signals

Short-horizon reversal is the contrarian hypothesis that recent losers tend to outperform recent winners over the next few sessions [1, 2]. For lookback k ,

$$\text{Rev}_{i,t}^{(k)} = - \left(\frac{C_{i,t}}{C_{i,t-k}} - 1 \right), \quad k \in \{1, 5\}. \quad (2)$$

A stock that lost 4% over the lookback receives a reversal signal of +0.04; a 4% winner receives −0.04. Larger values therefore identify recent losers. The third baseline signal is 12–1 momentum,

$$\text{Mom}_{i,t}^{(12,1)} = \frac{C_{i,t-21}}{C_{i,t-252}} - 1, \quad (3)$$

which omits the most recent month. The interaction specification adds the three pairwise products of cross-sectionally ranked baseline signals.

2.3 Technical library

Of the 101 formulaic alphas, 90 can be constructed from the course panel. Four formulas require subindustry, three require sector, and four reduce to identities. Two-digit SIC is used for the alphas defined at the industry level. The additional 15 signals cover intraday price shape, reversal and momentum at other lookbacks, volatility, unusual volume, Amihud illiquidity, and distance from the 52-week high. Appendix B records the complete implementation set.

3 Empirical Design

3.1 Model specifications

Table 1 defines the four predeclared comparisons. The technical models are competing specifications: their design matrices do not include the three baseline variables.

Table 1: Model specifications. Each is estimated separately at the one- and five-day horizons.

Label	Name	Predictors
A	Baseline	One-day reversal, five-day reversal, 12–1 momentum
B	Baseline + interactions	Baseline plus its three pairwise rank products
C	Pruned technicals	85 technical features retained by the frozen correlation filter
D	Full technicals	103 technical features surviving the coverage screen

The pruning sample is 2001–2010 and is excluded from subsequent model estimation. Features with less than 20% finite coverage are removed first, leaving 103 candidates. We then compute the Spearman correlation between each pair within every date and average those correlations through time. Starting with the pair having the largest absolute correlation, the feature with the larger mean absolute correlation to the remaining library is removed until $|\rho| < 0.7$ for every pair. This leaves 85 features.

Thresholds of 0.6 and 0.8 retain 76 and 92 features, respectively; the prediction target is absent from every step.

All features are converted each date to percentile ranks and linearly mapped to $(-1, 1]$. Missing ranks are assigned zero, the cross-sectional midpoint. This transformation makes coefficients comparable across formulas and limits the influence of extreme raw alpha values.

3.2 Elastic net and tuning

The elastic net combines variable selection with ridge shrinkage [4]. For each fold it solves

$$\min_{\beta_0, \beta} \frac{1}{2n} \sum_{j=1}^n (y_j - \beta_0 - x_j^\top \beta)^2 + \alpha \left[\lambda \|\beta\|_1 + \frac{1 - \lambda}{2} \|\beta\|_2^2 \right]. \quad (4)$$

The ℓ_1 component sets weak coefficients to zero; the ℓ_2 component stabilizes groups of correlated signals. This combination is suited to a technical library in which many columns share inputs and lookback windows.

The mixing parameter is selected from $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$. For each value, a training-derived path of 30 penalties runs from $\alpha_{\max}$, at which all slopes are zero, to $10^{-4}\alpha_{\max}$. Validation mean squared error selects (α, λ) . The selected pair is then refit on train plus validation. Figure 5 in the appendix shows that this procedure selects both sparse and dense models, including several all-zero folds.

3.3 Walk-forward validation and evaluation

For test year Y , training uses all model-sample years before $Y - 1$, validation uses $Y - 1$, and testing uses Y :

Train	Validation	Test
2011, ..., $Y - 2$	$Y - 1$	Y
fit path	select (α, λ)	evaluate once

The first test year is 2013. Five dates are purged from the end of both training and validation so that a five-day target cannot cross into the next block.

On each test date, rank IC is

$$\text{IC}_t = \text{Spearman}_i(\hat{y}_{i,t}, y_{i,t}). \quad (5)$$

The annual IC is the average over dates in that test year. If validation selects an all-zero model, every prediction is identical and IC is undefined. We report both the mean over active years and an all-fold mean that assigns zero to null folds.

Stocks are also sorted into ten equal-weight portfolios by predicted return. The 100%-gross long-short return is

$$r_t^{LS} = \frac{1}{2}(r_{10,t} - r_{1,t}). \quad (6)$$

One-day portfolios form daily. Five-day portfolios form on every fifth trading date and use $\sqrt{252/5}$ for annualization. Sharpe is calculated over the complete non-overlapping formation series, not as an average of annual Sharpes. Appendix E reports all five possible formation offsets. Returns are gross of costs.

4 Results

4.1 Overall comparison

Table 2 reports the complete scorecard: all four specifications at both horizons. Active IC conditions on a non-null fold; all-fold IC assigns zero to a null fold. The spread column is the mean return of

Equation (6) in basis points per formation period.

Table 2: Out-of-sample comparison, 2013–2025. Sharpe ratios use the full non-overlapping formation series and are gross of costs.

Specification	h	Active IC	Active	All-fold IC	Spread (bp)	Sharpe
Baseline	1d	0.0102	8/13	0.0063	0.27	0.08
Baseline	5d	0.0052	8/13	0.0032	−1.90	−0.12
Baseline + interactions	1d	0.0109	9/13	0.0076	0.41	0.13
Baseline + interactions	5d	0.0052	8/13	0.0032	−0.85	−0.05
Pruned technicals	1d	0.0079	10/13	0.0061	1.56	0.63
Pruned technicals	5d	0.0041	10/13	0.0032	1.78	0.12
Full technicals	1d	0.0081	10/13	0.0062	1.52	0.61
Full technicals	5d	0.0042	9/13	0.0029	1.67	0.11

The benchmark and interaction specifications deliver the largest active-year ICs at both horizons. Once null folds are included as zero-signal years, the one-day ICs are similar across the baseline and technical specifications. Adding baseline interactions produces only a small change. The technical models have lower rank correlation over active years but larger extreme-decile spreads. Thus the main distinction is between average ordering and tail separation rather than a uniform improvement in prediction.

Pruning has almost no effect on the scorecard. The pruned and full technical models have nearly identical IC, spread return, and Sharpe at each horizon. This is consistent with the role of elastic-net shrinkage: many features removed by the correlation filter receive little weight when they are left in the full design matrix.

4.2 Variation through time

Figure 1 displays each annual test result. Predictability is small relative to its variation through time, and the ranking of specifications changes across years. The crosses mark years in which validation selected an all-zero model. These null outcomes occur for both simple and high-dimensional specifications, rather than being concentrated in one model family.

The predeclared stress years reinforce the same pattern (Appendix D). The one-day pruned model is active with positive IC in 2020 and 2025 and null in 2022. Its five-day counterpart is active in all three years. Since each stress observation is a single test year, these comparisons describe model behavior rather than establish a separate regime effect.

4.3 Cross-sectional portfolio sorts

Figure 2 averages the ten decile returns first within each test year and then across active years. The shaded regions are 95% intervals based on variation in yearly means. One-day technical specifications separate the two tails more clearly than the baseline. At five days, the baseline ranking is almost flat and its highest predicted decile underperforms its lowest. This explains how a positive full-cross-section IC can coexist with a negative long-short spread.

4.4 Selected signals and normalization

The selected technical model changes materially across folds. Figure 3 shows the twelve largest average coefficient magnitudes in the pruned specification. Range, reversal, and price-shape signals recur, but their weights and active sets vary. The one-day 2021 fit uses all 85 features; the five-day 2023 and 2025 fits use 85 and 83. These dense fits sit beside years in which every coefficient is zero. The library therefore behaves as a time-varying combination rather than a stable small set of dominant formulas. As a preprocessing sensitivity check, the baseline, pruned, and full models were also estimated after

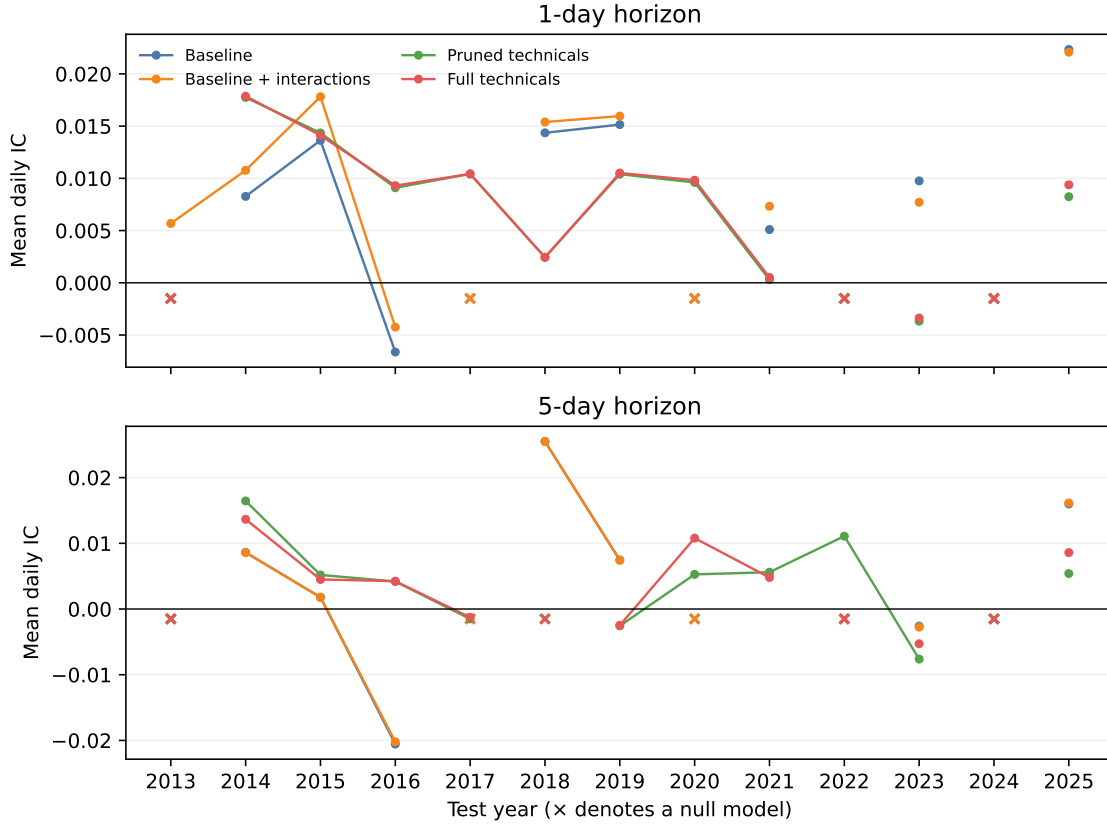


Figure 1: Mean daily cross-sectional Spearman IC by test year. Lines stop at null folds; crosses identify the corresponding years.

within-fold winsorization and standardization. Table 3 compares only years in which both versions are active. The similarity of the pruned and full models remains, while the sign of the scaling effect differs between the baseline and technical specifications. Rank normalization was the specified default; the ablation shows that the magnitude of the technical result is sensitive to feature scaling.

Table 3: One-day normalization ablation over paired active years. Δ is standardized minus ranked mean daily-Spearman IC.

Specification	Rank IC	Standardized IC	Δ	Paired years
Baseline	0.0087	0.0109	0.0022	5
Pruned technicals	0.0091	0.0035	-0.0057	7
Full technicals	0.0092	0.0034	-0.0058	7

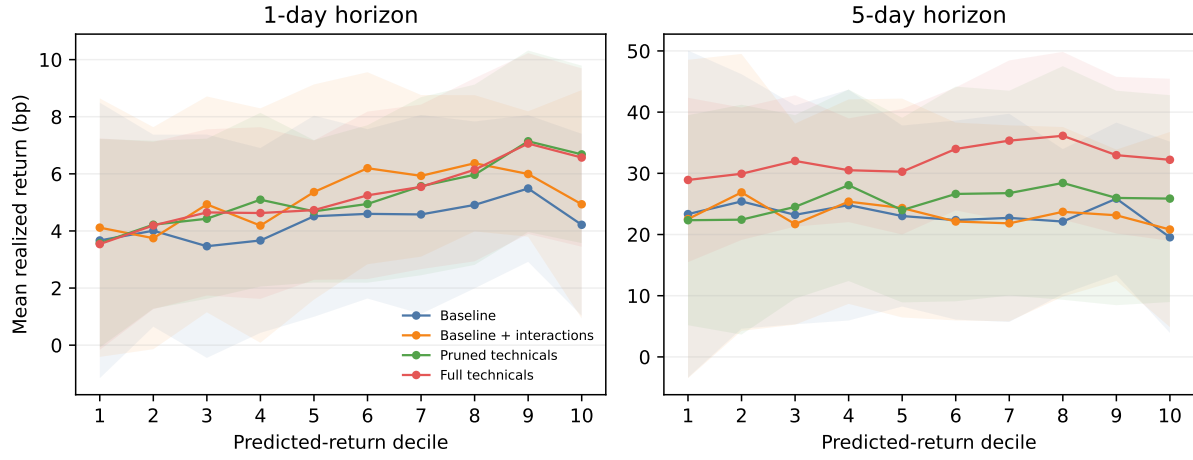


Figure 2: Mean realized return by predicted-return decile. Bands are ± 1.96 standard errors across annual means.

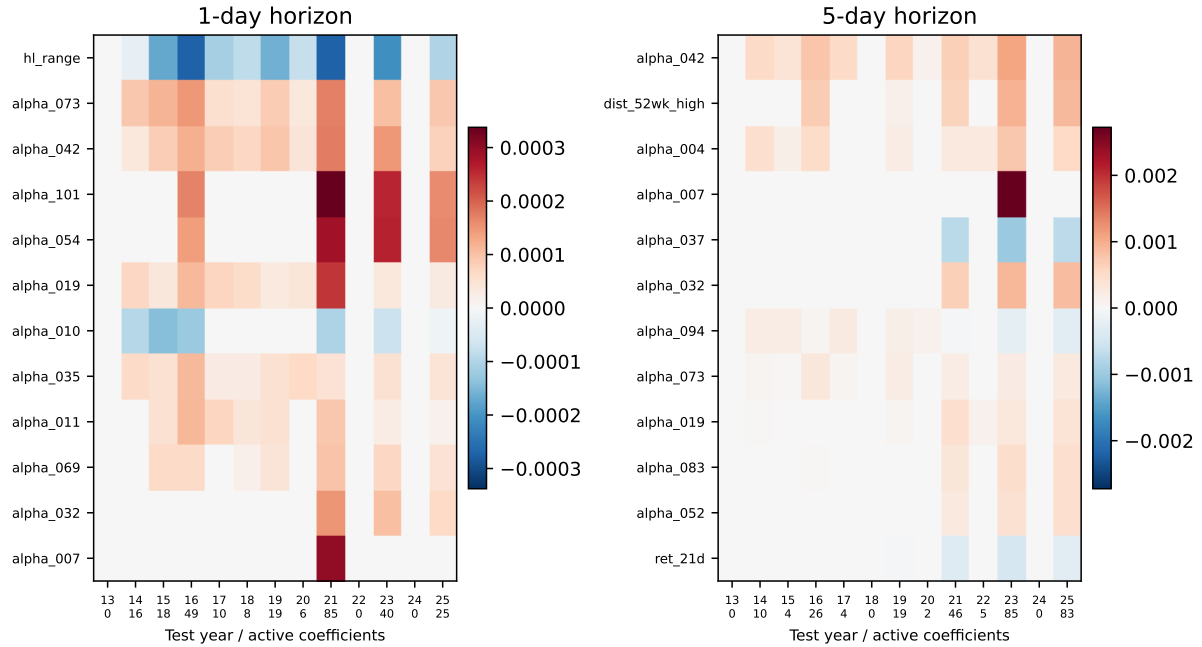


Figure 3: Elastic-net coefficients for the twelve largest average magnitudes in the pruned technical model. The second line of each x-axis label gives the number of active coefficients in that year.

5 Interpretation

The experiments give different answers for the center of the cross-section and its tails. Rank IC uses every security and rewards a broadly correct ordering. The portfolio sort uses only two deciles. A forecast may organize the middle of the cross-section while reversing its most extreme predictions, or produce modest average ordering with useful tail separation. The five-day baseline and one-day technical models illustrate these two cases.

The prune comparison is also informative. The frozen correlation rule reduces the library by 17% without materially changing out-of-sample results. It removes obvious redundancy and lowers computation, while elastic net performs the economically relevant target-aware selection inside each training fold. The coefficient heatmap suggests that selection instability is a more important feature of the experiment than the exact library size.

5.1 Limitations

First, Equation (1) is a split-adjusted price return. Compounded CRSP `ret` would include distributions and would follow the target construction suggested in the project brief. Second, several signals use the date- t closing bar while the target begins at that close. The portfolio interpretation therefore embeds a same-close execution convention; the IC comparison is more defensible than the simulated trading return. Third, transaction costs and turnover are absent. Daily decile rebalancing can make a statistically positive spread economically unusable.

Fourth, the supplied course panel contains roughly 1,000 names per day, but its point-in-time universe-membership construction cannot be independently verified from this repository. Active-year IC averages condition on validation selecting a nonconstant model. The all-fold column makes this conditioning visible, and the annual plot retains null and negative years. Five-day Sharpe also depends on the chosen non-overlapping formation offset; Appendix E reports the full range rather than selecting the best offset.

6 Conclusion

The broad technical specifications do not improve average cross-sectional rank IC relative to the reversal/momentum benchmark, although they generate greater one-day separation between the extreme deciles. Reducing the covered library from 103 to 85 features leaves performance essentially unchanged, consistent with elastic net absorbing much of the remaining redundancy. These weak and time-varying results support the use of simple benchmarks, yearly validation, and multiple evaluation metrics when studying technical signals.

References

- [1] N. Jegadeesh, “Evidence of Predictable Behavior of Security Returns,” *Journal of Finance*, 45(3), 881–898, 1990.
- [2] B. Lehmann, “Fads, Martingales, and Market Efficiency,” *Quarterly Journal of Economics*, 105(1), 1–28, 1990.
- [3] Z. Kakushadze, “101 Formulaic Alphas,” *Wilmott*, 2016(84), 72–81, 2016.
- [4] H. Zou and T. Hastie, “Regularization and Variable Selection via the Elastic Net,” *Journal of the Royal Statistical Society: Series B*, 67(2), 301–320, 2005.

A Data and feature audit

Table 4: Panel audit used before report preparation.

Check	Result
Rows after required-price filter	6,253,895
Unique securities / dates	3,253 / 6,266
Duplicate (<code>permno</code> , <code>date</code>) keys	0
Rows removed for missing close or market cap	1,180 (0.019%)
Missing open	0.069%
Listing segments / gaps over 10 days	8,748 / 5,495
One-day target missing	0.140%
Five-day target missing	0.699%

`alpha_007` illustrates the difference between coverage and variation. It is finite but equals -1 on 99.89% of rows and is cross-sectionally constant on 41.6% of dates. The feature remains in the frozen pruned list; elastic net decides whether it is active in each fold.

B Feature library

Skipped formulaic-alpha numbers are 48, 67, 90, and 100 (subindustry); 58, 76, and 79 (sector); and 59, 66, 82, and 89 (identities). The remaining 90 are implemented. The 15 classic features are `oc_ret`, `hl_range`, `overnight_gap`, `close_loc`, `ret_21d`, `mom_63`, `mom_6_1`, `vol_21d`, `parkinson_vol_21d`, `vol_of_vol_63d`, `vol_vs_mean21`, `dollar_vol_rank`, `amihud_21d`, `dist_52wk_high`, and `max_ret_21d`.

The correlation screen removes `alpha_096` and `alpha_097` for coverage, then retains 85 of 103 features at $|\rho| = 0.7$. Figure 4 displays the frozen-sample matrix.

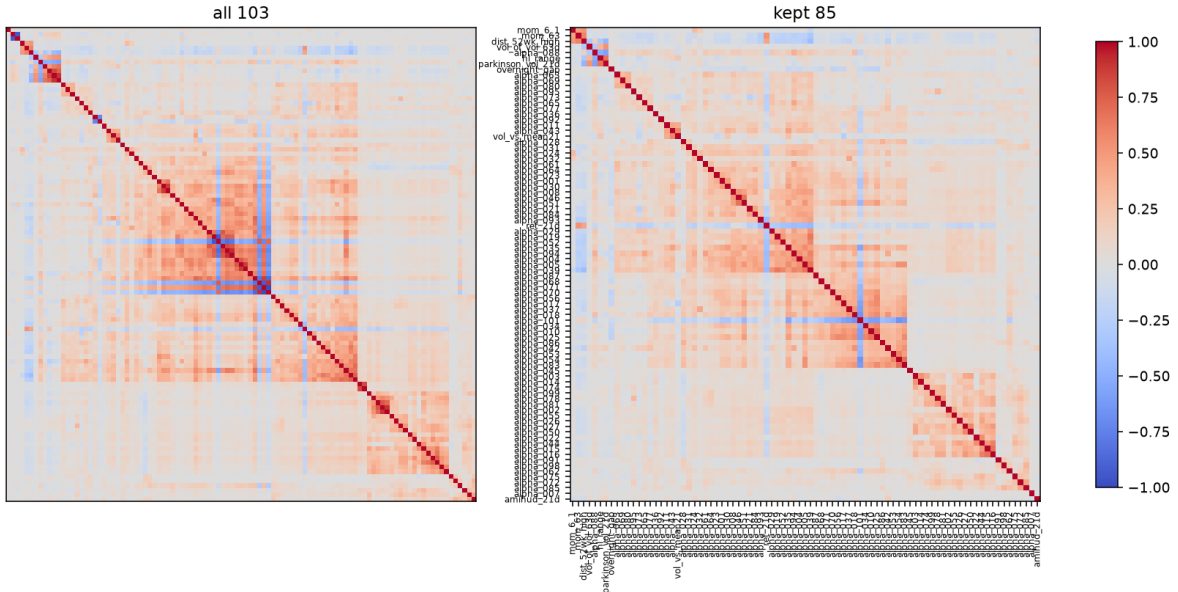


Figure 4: Mean daily Spearman correlations in the 2001–2010 pruning sample, before and after the $|\rho| = 0.7$ filter.

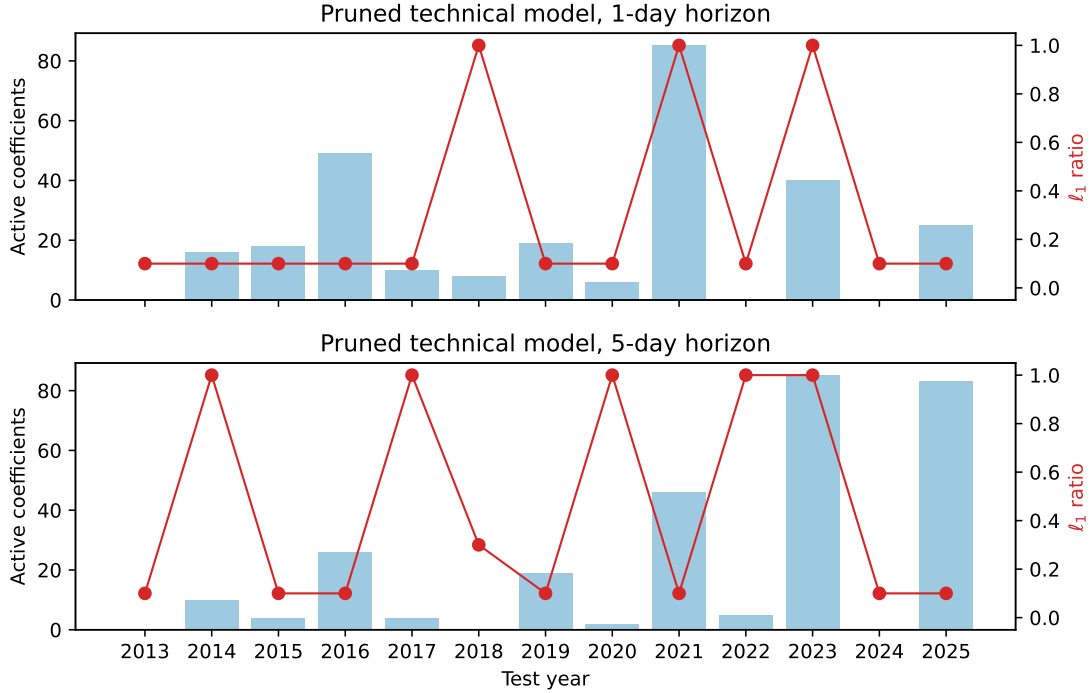


Figure 5: Selected ℓ_1 ratio and active-set size for the pruned technical model. Parameters are chosen on the preceding validation year.

C Tuning diagnostics

Raw α values are not directly comparable across folds because each path is scaled by its training-derived $\alpha_{\max}$. The active-set size shows the effective complexity. For the one-day pruned model, validation selects zero coefficients in 2013, 2022, and 2024 and all 85 in 2021. At the five-day horizon, 2013, 2018, and 2024 are null, while 2023 and 2025 retain 85 and 83 features.

D Stress-year results

Table 5: Predeclared stress years. Entries are annual mean daily IC and gross Sharpe. Dashes identify null folds.

Specification	h	2020		2022		2025	
		IC	SR	IC	SR	IC	SR
Baseline	1d	—	—	—	—	0.022	0.51
Baseline	5d	—	—	—	—	0.016	0.57
Pruned technicals	1d	0.010	0.72	—	—	0.008	0.25
Pruned technicals	5d	0.005	0.51	0.011	0.21	0.005	0.24

E Five-day formation-offset sensitivity

Table 6 summarizes, for each specification, the five possible non-overlapping five-day formation series.

Table 6: Five-day results across all non-overlapping formation offsets. Spread is the 100%-gross return in basis points per five-day period.

Specification	Min SR	Median SR	Max SR	Min spread	Max spread
Baseline	-0.15	-0.00	0.16	-2.51	2.81
Baseline + interactions	-0.16	-0.03	0.19	-2.68	3.39
Pruned technicals	0.12	0.38	0.72	1.78	11.86
Full technicals	0.11	0.34	0.64	1.67	10.95

F Reproducibility

The main experiment is run from the repository root with `uv run python run_models.py`. The normalization ablation behind Table 3 is reproduced by `uv run python run_normalization_ablation.py`, which refits the one-day models under both schemes and rewrites its summary CSV under `outputs/ablation_normalization/`. Report tables and figures are then generated by `uv run python run_report_figures.py`. The panel audit is reproduced by `uv run python run_panel_audit.py`. The small CSVs under `outputs/report/` support the model exhibits; the data-audit table comes from `outputs/panel_audit.json`. The course-provided raw panel and large prediction files are not versioned; full regeneration therefore requires access to the course data.